

# HAOZE SONG

Email: hzsong@cs.hku.hk ◊ Phone: +852 59890695 ◊ Website: <https://haozesong.github.io/>  
Room 414, Chow Yei Ching Building, The University of Hong Kong, Pokfulam, Hong Kong SAR

## RESEARCH INTEREST

---

Distributed Transactions, Fault-tolerance Consensus, and High-Performance Cloud Computing

## SUMMARY OF ACHIEVEMENTS

---

- Published 7 papers at top database and systems venues (including 3 first-author papers): **VLDB'25**, **SIGMOD'24**, **VLDBJ'24**, **SIGMOD'23**, **EuroSys'21**, **Performance'21**, and **TASE'19**.
- Another 6 papers are currently under submission, including 3 papers undergoing major revision: **SIGMOD'26 (first authored)**, **FAST'26**, and **TKDE'26**.
- Two system prototypes have been integrated into *real-world industrial products*:
  - K2, a multi-region transaction protocol, has been adopted by the Global TKV team, steadily evolving into the data backbone for next-generation cloud-native root services.
  - METIS, an HTAP-native query processing engine, has been integrated into Alibaba Cloud's flagship database product, PolarDB, as a component of its HTAP optimizer.

## EDUCATION

---

**The University of Hong Kong (HKU)**

Sep 2020 - Aug 2025 (Expected)

Ph.D. in Computer Science

Advisor: Dr. Heming Cui

Thesis: Achieving Efficient and Reliable Transactional/Analytical Processing for Geo-Distributed Clouds.

**University of Science and Technology of China (USTC)**

Aug 2016 - Jun 2020

B.S. in Computer Science and Engineering

Huaxia Yingcai College (Special Elite Class)

GPA: 3.81/4.3

Rank: 7/153 (Top 5%)

Outstanding Graduates Honor

**Shanghai Jiao Tong University (SJTU)**

Jun 2018 - Sept. 2018

Exchange Student in Computer Science

GPA: 4.3/4.3

## EXPERIENCE

---

**DAMO Academy, Alibaba**

June 2022 - Sept. 2023

Database Group on System and Kernel Research

Advisor: Wenchao Zhou and Feifei Li

Research Internship

- Project: Cloud-native HTAP, which achieves  $100\times$  speed up for TPC-H queries and  $\leq 5ms$  visibility delays for data replication. The new technologies (i.e., efficient data replication and hybrid query plans) have been integrated into PolarDB (the core database product) in Alibaba Cloud.

## PUBLICATIONS

---

[C1] Haoze Song, Yongqi Wang, Xusheng Chen, Hao Feng, Yazhi Feng, Xieyun Fang, Heming Cui, and Linghe Kong. K2: On Optimizing Distributed Transactions in a Multi-region Data Store with True-Time Clock. *International Conference on Very Large Data Bases (VLDB)*, 2025

[C2] Haoze Song, Wenchao Zhou, Feifei Li, Xiang Peng, Heming Cui. Rethink Query Optimization in HTAP Databases. *ACM International Conference on Management of Data (SIGMOD)*, 2024

[J1] Haoze Song, Wenchao Zhou, Heming Cui, Xiang Peng, Feifei Li. A Survey on Hybrid Transactional and Analytical Processing. *International Journal on Very Large Databases (VLDBJ)*, 2024

- [C3] Jianying Wang, TongLiang Li, Haoze Song, Xinjun Yang, Wenchao Zhou, Feifei Li, et al. *PolarDB-IMCI: A Cloud-Native HTAP Database System at Alibaba*. *ACM International Conference on Management of Data (SIGMOD)*, 2023
- [C4] Xusheng Chen, Haoze Song, Jianyu Jiang, Chaoyi Ruan, Cheng Li, Seng Wang, Gong Zhang, Reynold Cheng, Heming Cui. *Achieving Low Tail-latency and High Scalability for Serializable Transactions in Edge Computing*. *The European Conference on Computer Systems (EuroSys)*, 2021
- [C5] Xusheng Chen, Shixiong Zhao, Ji Qi, Jianyu Jiang, Haoze Song, Cheng Wang, Tsz On Li, Hubert Chan, Fengwei Zhang, Xiapu Luo, Sen Wang, Gong Zhang, Heming Cui. *Efficient and DoS-resistant Consensus for Permissioned Blockchains*. *ACM SIGMETRICS Performance*, 2021
- [C6] Yu Zhang, Haowei Deng, Quanxi Li, Haoze Song and Leihai Nie, *Optimizing quantum programs against decoherence: delaying qubits into quantum superposition*. *International Symposium on Theoretical Aspects of Software Engineering (TASE)*, 2019

## PAPERS IN SUBMISSION (PREPRINTS)

---

- [S1] Haoze Song, Xusheng Chen, Ruijie Gong, Tianxiang Shen, Cheng Li, Hao Feng, Sen Wang, Gong Zhang, Heming Cui. *Perseus: Achieving Strong Consistency and High Data Freshness for Scalable Geo-distributed HTAP*. *Major Revision in SIGMOD*, 2026
- [S2] Haoze Song, Ruijie Gong, Xusheng Chen, Tianxiang Shen, Yuhao Qing, Sen Wang, Gong Zhang, Hao Feng, and Heming Cui. *RELAY: High-performance Transactions in Heterogeneous Networks via Consistency Tiering*. Submitted to *Transactions on Computer Systems (TOCS)*
- [S3] Tianxiang Shen, Ji Qi, Haoze Song, Gong Zhang, Xiaopu Luo, and Heming Cui. *Achieving Efficient and Compressible Indexing on Encrypted Databases*. *Major Revision in TKDE*
- [S4] Guoli Wei, YongKun Li, Haoze Song, Lulu Yao, Yinglong Xu, Bokang Zhang, Liu Tang, and Qiu Cui, *DMTree: Resolving the Performance Tradeoffs of Tree Indexing on Disaggregated Memory*. *Major Revision in USENIX Conference on File and Storage Technologies (FAST)*, 2026
- [S5] Zheng Liu, Haoze Song, YongKun Li, Yinglong Xu, Patrick P.C. Lee, Xusheng Chen, Yazhi Feng, and Hao Feng. *PartialKV: On Optimizing Partial Access for Persistent Key-Value Store in Modern Memory Hierarchy*.
- [S6] Zekai Sun, Xiuxian Guan, Haoze Song, Yuhao Qing, Tianxiang Shen, Dong Huang, Fangming Liu, Heming Cui. *Hybrid-Parallel: Achieving High Performance and Energy Efficient Distributed Inference on Edge Computing*. Submitted to *International Conference on Mobile Computing and Networking (MobiCom)*, 2025

## TECHNOLOGY TRANSFER (PATENTS)

---

- [P1] Haoze Song, Xusheng Chen, Yazhi Feng, Xieyun Fang, *A scalable high-performance and high-precision timestamp batching framework*. Ref.92083535
- [P2] Haoze Song, Yongqi Wang, Xusheng Chen, Yazhi Feng, *An efficient, strongly consistent, and decentralized visibility control method based on high-precision clocks*. Ref. 92076896
- [P3] Xusheng Chen, Haoze Song, Jianyu Jiang, Heming Cui, Sen Wang, Peng Wang, and Gong Zhang. *A system in achieving low tail-latency and high scalability for serializable transactions in edge computing*. CN 2021101523346.3

## PROFESSIONAL SERVICE

---

External/Artifact reviewers in system conference: *OSDI*, *NSDI*, *ATC*, *EuroSys*, and *DSN*

External reviewers in system journals: *IEEE Transactions on Parallel and Distributed Systems (TPDS)*

**TEACHING EXPERIENCE**

---

COMP-3358 Distributed and Parallel Computing, HKU, Teaching Assistant 2023 Spring, 2024 Spring  
COMP-7305 Cluster and cloud computing, HKU, Teaching Assistant 2021 Summer, 2022 Spring

**SELECTED AWARDS**

---

Outstanding Research Intern, Alibaba Group, Top 5% 2023  
Postgraduate Scholarships, The University of Hong Kong 2020~2025  
Outstanding Graduates Honour, University of Science and Technology of China, Top 10% 2020  
Elite Class Honor, University of Science and Technology of China, Top 10% 2018, 2019, 2020  
Second Class scholarship, University of Science and Technology of China, Top 20% 2019, 2020